

**RIVET LOADS IN PLANE LOADS WITH
ECCENTRIC LOADS**

Section: **3.2 – MECHANICALLY FASTENED
JOINTS**

Chapter: **3213**

Title: **RIVET LOADS IN PLANE LOADS WITH
ECCENTRIC LOADS**

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1 INTRODUCTION

1.1 OBJECT

This chapter defines the methodology to use for obtaining the loads in the rivets or bolts, which make up the mechanically fastened attachment between a fitting and a support structure, when ALL the fasteners are installed in just one plane, and the utilization of other more sophisticated methods (like numerical FEM analysis) are not appropriate.

1.2 SCOPE AND LIMITATIONS

Calculation of the loads in the rivets attaching a fitting to a structure, when the rivets distribution is enclosed within just one plane.

- ✓ Fitting assumed stiff as a rigid body
- ✓ Linear elastic relationships between displacement and forces at rivets nodes

The stress engineering should take special care if multiple load paths exist e.g. if the loaded part is joined to more than one part. In this case the individual stiffness/compliance of each part may significantly alter the load path compared to the assumed rigid body solution.

This method is not recommended for final certification analysis of fastened eccentric joints with out-of-plate loads unless the analysis parameters (the rivets stiffness, especially in tension) for the case of study, or for a reasonably similar configuration, were previously validated by precise means (FEM numerical simulation or tests).

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2 METHOD DESCRIPTION

The non-numerical method proposed herein is supported in two main assumptions: the fitting behaves as a rigid-body, and the point-stiffness at each rivet position is known. The first assumption generally fits well with the behaviour of real well designed parts, i.e. fittings made of machined metallic alloys, with sound robust loads paths from load application points to attachment rivets. The estimation of the stiffness of each rivet, in particular along the direction of its axis (stiffness in tension) is a more complex topic, dependent on the rivet itself, on the flange of the fitting where it is installed, and on the supporting structure. Expressions for estimating rivets stiffness were presented in previous chapters of this manual (see chapter 3210 "CALCULATION OF RIVETS LOADS").

Two methods for structural details of this type are considered, function of the type of loads set applied on the bolted joint, but it is just for illustrative purposes. The method proposed for the general case is applicable to all kinds of applied loads.

2.1 IN-PLANE JOINT WITH ECCENTRIC IN-PLANE LOADS

For this case not only all the attachment rivets of the fitting are installed in one plane, but also the loads set that has to be sustained by the joint is contained in that plane.

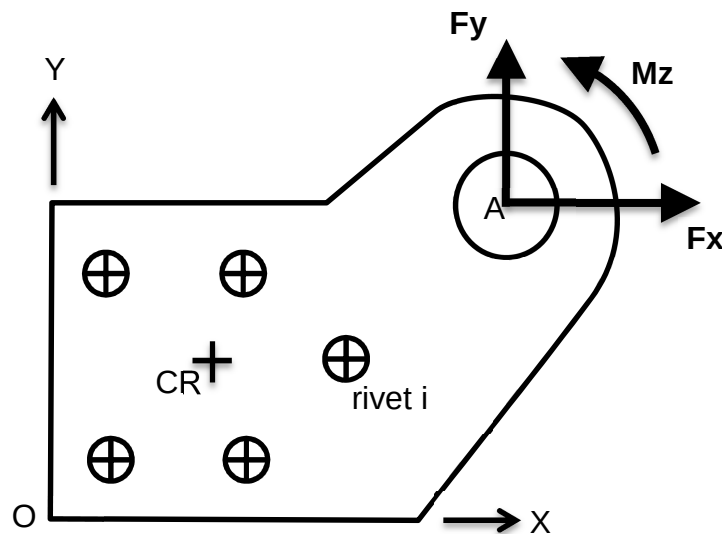


Figure 1. In-plane fastened joint submitted to eccentric load in the plane

This case has a comprehensive solution in classical stress literature, as Bruhn's chapter D1.17 "Multiple Riveted or Bolted Joints Subjected to Eccentric Loads", or Niu's chapter 9.4 "Eccentric Joints". That approach is here below extended just to consider different rivets stiffnesses within the joint.

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Centre of riveting

First of all, the “centre of riveting” is calculated as follows:

- ✓ Weight $W = \sum K_i$
- ✓ Coordinates $X_{cr} = \sum (K_i \cdot X_i) / W$
 $Y_{cr} = \sum (K_i \cdot Y_i) / W$

Change of axis system

Problem inputs are translated to a reference system with origin at the centre of riveting:

- ✓ Rivets coordinates $X_i' = X_i - X_{cr}$
 $Y_i' = Y_i - Y_{cr}$
- ✓ Applied moment $M = M_z + (X_A - X_{cr}) \cdot F_y - (Y_A - Y_{cr}) \cdot F_x$

The “Inertia” of the joint vs applied moments:

- ✓ Inertia $I = \sum (K_i \cdot R_i^2) = \sum [K_i \cdot (X_i'^2 + Y_i'^2)]$

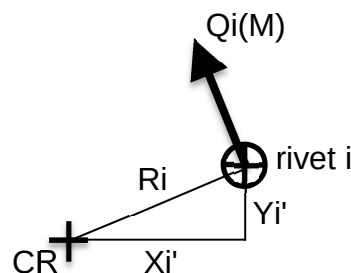


Figure 2. Shear load at rivet due to applied moment

Shear loads in rivets

- ... due to applied forces: $Q_{ix}(F_x) = F_x \cdot K_i / W$
 $Q_{iy}(F_y) = F_y \cdot K_i / W$
- ... due to applied moment: $Q_i(M) = M \cdot K_i \cdot R_i / I$
 $Q_{ix}(M) = - Q_i(M) \cdot Y_i' / R_i$
 $Q_{iy}(M) = Q_i(M) \cdot X_i' / R_i$
- ... total per rivet: $Q_{ix} = Q_{ix}(F_x) + Q_{ix}(M)$
 $Q_{iy} = Q_{iy}(F_y) + Q_{iy}(M)$
 $Q_i = (Q_{ix}^2 + Q_{iy}^2)^{0.5}$

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2.2 IN-PLANE JOINT WITH ECCENTRIC LOADS (GENERAL CASE)

The general case, including in-plane and out-of-plane loads, can be decomposed in two independent/uncoupled analyses:

- ✓ The first problem shall include only the components of the applied loads embedded in the joint plane (i.e. F_x , F_y and M_z , with the plane joint the XY-plane). For this case the methodology was described in previous sub-chapter. It will cause just shear loads in the fasteners
- ✓ The second case shall include only the components of the applied loads not contained in the joint plane (i.e. F_z , M_x and M_y).

This problem causes tension forces in the fasteners and contact forces between the attached plates. In other words, at some fitting locations fasteners will react applied forces as tension forces, whereas at other locations the fitting base plate will be pushing the supporting plate (there cannot be “compression” forces in the rivets!). A dedicated methodology for approaching this problem is explained in HSB 21030-01 (analogy with beam section subjected to axial loads and bending moments). However, this problem is highly non-linear. Alternative methods were conceived to deal with it and find a comprehensive solution, e.g. that described per Ref. 4 and Ref. 5. Anyway, for the time being, none of those are considered as Airbus DS harmonized methods.

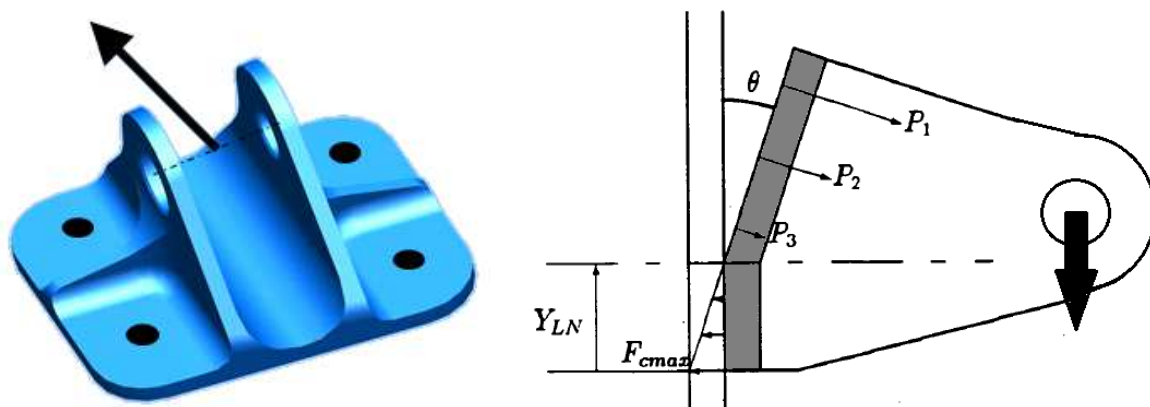


Figure 1. In-plane fastened joint submitted to eccentric load with load out of the plane

The analysis of this type of problem shall be considered a special case of the more general case of a fitting attached to several support planes, as described in chapter §3214 “RIVETS LOADS IN ECCENTRIC 3D JOINTS”. The same method proposed in that chapter to calculate rivets loads in 3D joints shall be used for planar joints.

Provided the robustness of the fitting and support structure are able to react calculated tension in bolts and contact forces between plates, and that areas in with significant contact forces are clearly identified, the “dummy fastener” approach as suggested by the HSB 21030-01 (paragraph 4.7) may be used. This strategy reduces calculated tension loads in bolts and, consequently, increases resulting static margins. It must be remarked that it is not recommended for initial sizing phases (characterized by sufficient design freedom and many uncertainties), and, on the other hand, prior to present a structural justification supported in this approach for final certification analysis, further validation via numerical simulation or tests, on the specific configuration or on a very similar one, is strongly recommended.

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3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
FEM	-	Finite Element Model
O	-	Origin of reference axis system for analysis
A	-	Loads application point (coordinates XA and YA)
X_i, Y_i	-	Coordinates of rivet i
K_i	N/mm / -	Stiffness/effectiveness of rivet i (all values must be absolute/dimensional or relative/non-dimensional)
X_{cr}, Y_{cr}	mm	Coordinates of Centre of Riveting (CR)
X'_i, Y'_i	mm	Coordinates of rivet i relative to the CR
R_i	mm	Distance from the CR to the rivet i
W	N/mm / -	Weight of the joint vs applied forces (same dimension as rivets stiffness/effectiveness)
I	Nmm / mm ²	Inertia of the riveted joint vs applied moments (different units if rivets stiffness are given as non-dimension effectiveness)
F_x	N	X-force applied at point A
F_y	N	Y-force applied at point A
M_z	Nmm	Z-moment applied at point A
M	Nmm	Applied moment relative to the CR
Q_{ix}, Q_{iy}, Q_i	N	Components and resultant shear load in rivet i

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 2	[M. C. Y. Niu]	2 nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 3	HSB21030-01	Is. D	Internal load distribution of fastener groups
Ref. 4	MANUAL DE DISEÑO CASA	01/95	"DISTRIBUCION DE CARGAS EN UNIONES REMACHADAS" (Vol. 8 – Chapter 064)
Ref. 5	NT-T-ADCE-021/92	-	Manual Teórico del Programa BULONES, version 2.0

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SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures (Module BOLTS3D)

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
1.0 29/04/2020	G. Baños	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Methods (TEASP-TL3)	G. Baños 04/12/2018	F. Glock 29/04/2020

Table 3. *Approval signatures table for last revision of this chapter*